

Computing A Haar Feature

 H_A

White = 1, Black

Response to Filter H_A at location (i, j) :

$$V_A[i, j] = \sum_m \sum_n I[m, n] H_A[m - i, n - j]$$

$$V_A[i, j] = \sum (\text{pixel intensities in white area}) \\ - \sum (\text{pixels intensities in black area})$$